



SUPERMARKET SALES DATA INSIGHTS

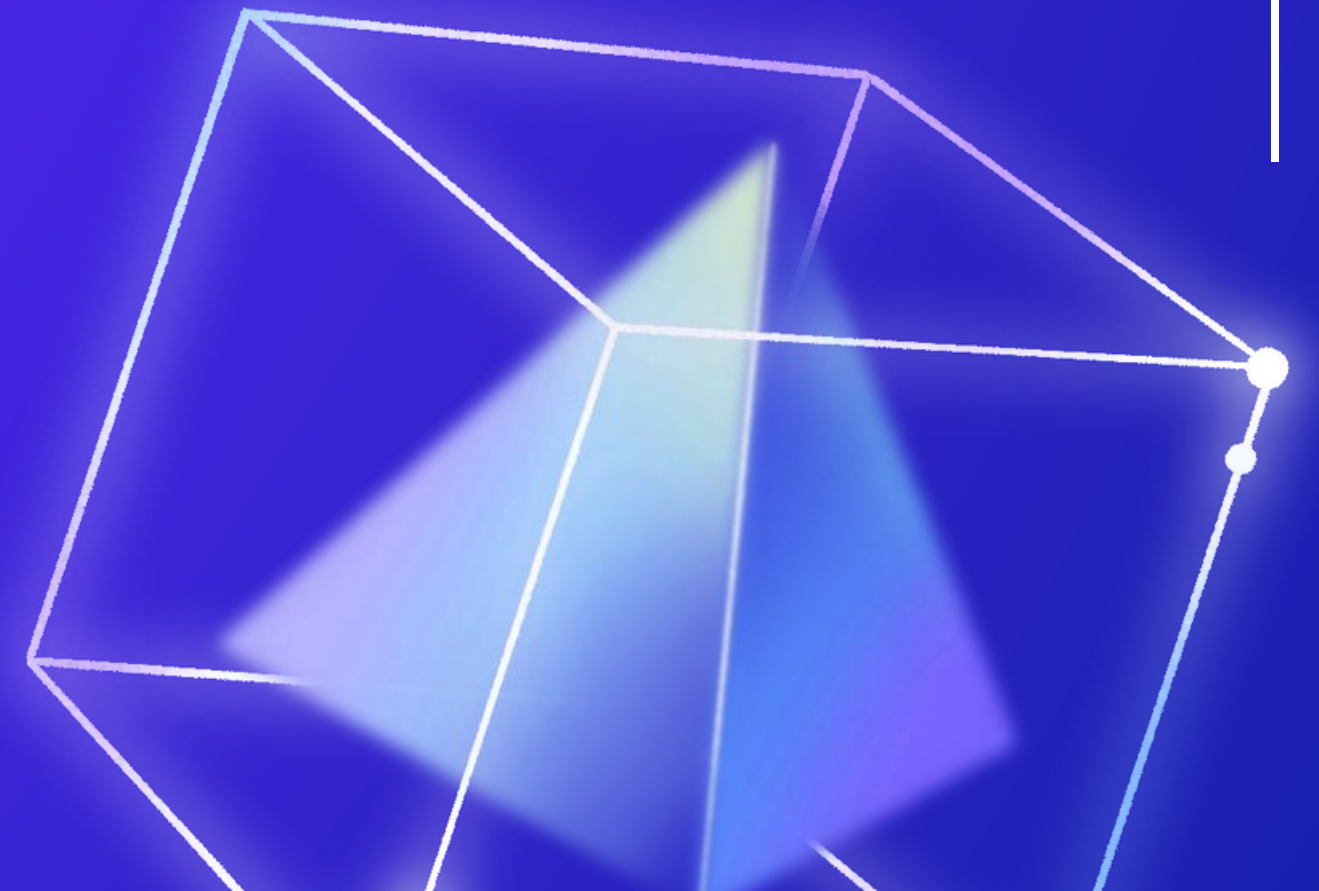
CAPSTONE PROJECT

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RAW DATASET

Dataset

Supermarket Sales Sample Data

Sumber

LINK: <https://www.exceldemy.com/learn-excel/sample-data/#:~:text=Supermarket%20Sales%20Sample%20Data%20in%20Excel>

Field Utama

Order No.

Order Date

Customer Name

Ship Date

Retail Price

Order Quantity

Tax

Total



PROJECT OVERVIEW

Latar Belakang

Supermarket memiliki transaksi harian yang besar dan kompleks. Analisis data diperlukan untuk mendukung pengambilan keputusan yang lebih baik.

Tujuan Proyek

- Analisis pola penjualan dan tren.
- Identifikasi high value orders.
- Memberikan insight dan rekomendasi strategis.
- Menggunakan AI (IBM Granite Model via Replicate) untuk meningkatkan analisis.



ANALYSIS PROCESS



Data Preparation:

- Import
- Cleaning



Exploratory Data Analysis (EDA):

- Tren bulanan
- Distribusi order
- Revenue analysis



AI Support:

- Classification → High vs Low Value
- Summarization → Ringkasan insight
- Recommendation → Strategi bisnis

EXPLORATORY DATA ANALYSIS

CONTINUOUS LEARNING STATISTIK DESKRIPTIF (MEAN, MEDIAN, TOTAL)

TREN PENJUALAN BULANAN

DISTRIBUSI ORDER QUANTITY

KONTRIBUSI REVENUE

Statistik Data				
	Order No	Order Date	Ship Date	Retail Price (USD)
count	70.000000	70.000000	70.000000	70.000000
mean	1035.500000	45309.000000	45311.900000	84.418571
std	20.351085	10.175276	10.003115	57.043988
min	1001.000000	45292.000000	45294.000000	19.990000
25%	1018.250000	45300.250000	45304.250000	39.990000
50%	1035.500000	45309.000000	45310.500000	69.990000
75%	1052.750000	45317.750000	45320.000000	129.990000
max	1070.000000	45326.000000	45330.000000	199.990000

	Order Quantity	Tax (USD)	Total (USD)
count	70.000000	70.000000	70.000000
mean	2.400000	14.097600	155.073600
std	1.334058	3.894425	42.838676
min	1.000000	2.999000	32.989000
25%	1.000000	12.247500	134.722500
50%	2.000000	14.995000	164.945000
75%	3.000000	14.999000	164.989000
max	5.000000	29.997000	329.967000

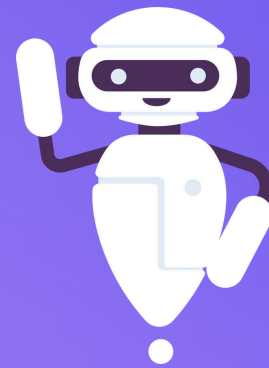




INSIGHT & FINDINGS

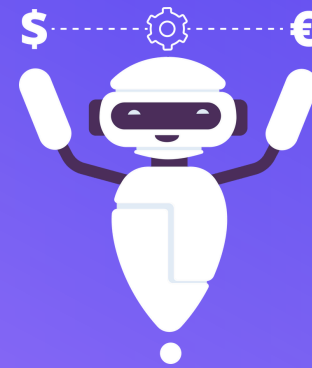
- Pareto Principle: 80% revenue berasal dari 20% transaksi besar.
- Terdapat seasonality trend pada order quantity.
- Produk dengan harga tinggi → margin besar, meski volume rendah.

AI SUPPORT



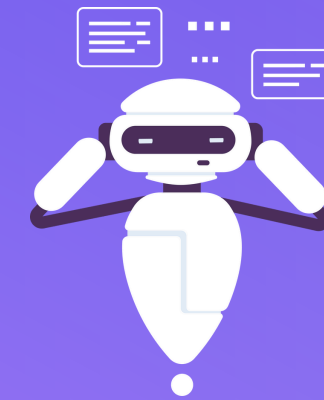
MODEL

Granite-3.0-8b-instruct (IBM)
via Replicate API.



PENGUNAAN

- Klasifikasi transaksi high vs low value.
- Ringkasan insight dataset.
- Rekomendasi strategi penjualan.



CONTOH PROMPT

- "Classify transactions as High Value or Low Value"
- "Summarize key insights from dataset analysis"
- "Recommend 3 actionable strategies to improve revenue"

CONCLUSION & RECOMMENDATION

01

- High value orders memegang kontribusi utama terhadap revenue.
- Ditemukan pola musiman pada penjualan.
- Shipping gap memerlukan perhatian manajemen operasional.

02

- Fokus promosi pada high-value products.
- Terapkan strategi diskon/off-season promo.
- Perbaiki manajemen proses shipping.

CLOSING

Kesimpulan Umum:

- Data analytics + AI memberikan insight strategis bagi supermarket.

Tools yang digunakan:

- Python (Pandas, Matplotlib)
- Google Colab
- IBM Granite via Replicate API



THANK YOU!

